Paul Thielen

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Education

Carnegie Mellon University (CMU) - Pittsburgh PA *May 2027*

Master of Science in Electrical and Computer Engineering (AI/ML Concentration) | GPA: **4.00/4.00**

- Coursework: Computer Vision, Estimation Detection and Learning, Intro to Machine Learning

Technische Universiteit Eindhoven (TUE) – Eindhoven, NL *Jun 2024*

Bachelor of Science in Data Science | GPA: **3.94/4.00** (Cum Laude)

- Minor: Cognitive Science & Artificial Intelligence

Internship Experience

NBA - New York *June 2026 - Aug 2026*

Enterprise AI Intern

- Built and deployed a full-stack Nuxt.js app with an LLM-based classification pipeline that centralized 500+ internal posts into a searchable catalog **adopted by 25+ stakeholders** across two business units
- Developed a **RAG**-powered Snowflake agent over the catalog, enabling non-technical teams to run self-serve analysis
- Automated go-to-market briefs with a Copilot agent that generates talking points, cutting prep time by **40%**
- Engineered 4 document-analysis AI agents, each tailored to a business vertical, that turn unstructured documents into structured insights for internal employees, saving an average of **7 minutes** of review time per document

Uber - Amsterdam, Netherlands *Jan 2023 – Jul 2023*

Data Scientist Intern

- Developed and implemented a **SARIMA model** in Python to project sales quota attainments, enabling the sales team to refine strategies and achieve targets
- Applied time series analysis and predictive modeling techniques in **Python** to identify high-impact sales strategies and improve decision-making, resulting in an **18%** increase in weekly sales income
- Wrote automated **PrestoSQL** queries on large spatiotemporal data, reducing manual workload by **16 hours per week**

Research Project Experience

Medical Data Under Shadow Attacks via Hybrid Model Inversion – AISTATS *Jun 2024 – May 2025*

- Developed MEDUSA, a hybrid gray-box **model inversion** framework combining gradient-based optimization with inverse modeling in **PyTorch**, in collaboration with a *Stanford AIMI* visiting researcher
- MEDUSA improved medical image reconstruction fidelity by up to **12%** over gradient-based reconstructions
- Demonstrated use of reconstructed medical images as synthetic training samples, attaining training performance within **5%** of real data benchmarks
- Invented and evaluated the KNN Smearing Defense method, reducing reconstruction quality by up to **64%**

Academic Project Experience

Food Insecurity in South Sudan - TUE Data Challenge *Feb 2024 – Apr 2024*

- Led a team project in collaboration with the *Zero Hunger Lab* to predict food insecurity risks in South Sudan using text data from news sources and climate data, earning a **9/10 grade**
- Trained **linear regression** models in **Sklearn** on climate data and GPT-3-extracted mentions of violence from local and global news articles about South Sudan to forecast monthly food insecurity risk scores
- Modeled impact of the Bentiu takeover using a **Bayesian Structural Time Series** model, identifying a **71%** statistically significant increase in food insecurity risk scores and a persistent post-conflict distribution shift

Skills

Programming Languages:	Python, R, SQL, MATLAB, TypeScript, JavaScript
Deep Learning Frameworks:	PyTorch, TensorFlow, JAX, Keras, Hugging Face
Machine Learning & Data Science Tools:	Sklearn, Numpy, Pandas, Spark, Seaborn, OpenCV, PowerBI, Tableau, Streamlit
Cloud Computing Technologies:	Microsoft Azure, Amazon Web Services, Google Cloud Platform, Snowflake

Certificates

Microsoft AI & ML Engineering (Coursera)	*Mar 2026*	Machine Learning Track (DataCamp)	*Aug 2024*
Microsoft Excel (Coursera)	*Jan 2025*	Deep Learning Track (DataCamp)	*Jul 2024*